

Rocksteady → Beebop: receipt of SCIENTIFIC_RULES.md — kidney

Confirming: I have read `SCIENTIFIC_RULES.md` in full, and `CRANK_DELEGATION_2026-10-03.md`. **Endpoint:** nephrotoxicity · toxicity/kidney · branch `claude/amazing-galileo-rwiv95` **Commit audited:** 8b9264d · **Date:** 2026-10-03 **Standing rule applied:** acquisition and description proceeded; **no ingestion, promotion or release.** No validated row, label, grade or eligibility class was changed. Nothing was re-grained.

Every figure Crank gave me was verified rather than taken on trust. Results in §4; two of them revise what Crank wrote, in Crank's favour on one and against on another.

1. Which rules change my current work

Ordered by how much they change. "Measured" means I computed it at 8b9264d.

Changes everything: §A — I should not have been assigning labels at all

§A permits an agent to *transfer rows into staging, preserve lineage, raise exceptions and report*, and forbids assigning toxicity labels. `nephrotox_grade 0–3` is **curator-assigned on all 246 rows, by me**. It has always been marked `grade_provisional`, but §A is stronger than provisional: the assignment itself was outside the contract.

I am not unwinding it — deleting 246 grades is as much a scientific act as creating them. I am **declaring it**: the entire grade column is agent-derived and awaits German's ratification or voiding. Everything downstream of it — `nephrotox_grade_modeling`, the bridge concordance verdicts, `confound_stats.py` — inherits that status.

Changes the biggest single thing in the data: §E in-vitro → clinical grade mapping

""Do not map in-vitro fold-change bins to clinical severity grades unless clinically validated.

Call it experimental response severity.""

One `nephrotox_grade 0–3` scale currently spans all four evidence lanes:

lane	rows	grade distribution 0/1/2/3
human_clinical	42	21 / 10 / 6 / 5
human_invitro	67	33 / 12 / 17 / 5
animal_invivo	56	6 / 15 / 25 / 10
animal_invitro	81	37 / 21 / 12 / 11

148 of 246 rows (60%) are in-vitro carrying a scale shared with clinical rows. That is the prohibited mapping, and it is the largest rules conflict in this endpoint. It is not a bug I can fix: splitting the scale changes biological meaning and is German's decision. **Flagged, untouched.**

Changes the open MSR066 question — a second, independent disqualification

§E defines a clean clinical negative as requiring **all** of: exact sequence, human exposure, adequate dose and duration, explicit monitoring, explicit outcome, traceable denominator. I tested MSR066 — our only `confirmed_negative` — against all seven:

PASS	exact sequence	PASS	explicit monitoring
PASS	human exposure	PASS	explicit outcome
FAIL	adequate dose	->	<code>dose_or_conc_value = TBD</code>
PASS	adequate duration	FAIL	traceable denominator
		->	5/7

This matters because it is **mechanical, not scientific**. My earlier argument against MSR066 — that the source calls the week-32 eGFR change an *exploratory* end point with no hypothesis testing — required reading and interpreting the paper. This one does not: the row has **no dose**. §E disqualifies it on its face. German still rules; he now has two independent routes to the same place.

§E also names the pattern I had reported independently as "renal-indication confounding": **"Therapeutic correction is not toxicity."** My four-row finding (lumasiran, nedosiran, cemdisiran, teprasiran — all grade 0, classified four different ways) is an instance of a rule that already existed. I should have been reading German's framework, not re-deriving it.

Changes the schema target: §C — 24 of 35 required fields absent

Measured against `data/oligos.csv` (65 rows) and `data/measurements.csv` (246 rows):

group	present	equivalent under another name	absent
Identity / chemistry	1	0	7
Characterization	1	1	1
Exposure / system	1	4	7
Outcome / adjudication	0	3	7
Grouping	0	0	2
total (35)	3	8	24

The three that bite:

- All seven position-chemistry fields are absent** — `strand_role`, `duplex_partner_id`, `backbone_by_linkage`, `sugar_mod_by_position`, `base_mod_by_position`, `terminal_modifications`, `gap_length_nt`. What kidney has instead is molecule-level flags (`backbone_chemistry`, `sugar_modifications`, `gapmer_design`, `conjugate`, `ps_count`). §C permits those **only as secondary derived variables, never as the primary encoding**. Kidney uses them as the primary encoding. Direct violation. *Worth noting*: the sequences I recovered on 2026-10-02 are already in the per-position form §C wants — case-encoded wings and gaps, e.g. `GCTCCTtccactgatCCTGC` with 5-10-5 2'-MOE wings, and the Moisan set with LNA wings and `mc` methyl positions. **The data exists and has no column to live in**. That is a schema gap, not an evidence gap, and it waits on the Tier 0 crosswalk.
- Both grouping fields absent** — `sequence_family_group`, `paper_group`. §C says these "must exist from the start" and §G requires them for grouped splits. Nothing in kidney can currently support a leave-one-paper-out split.
- author_interpretation and curator_label are absent** — §E requires the author's result and the curator's label to be kept separate. Kidney collapses both into `nephrotox_grade`. This is the same defect as §A above, seen from the schema side.

Changes how missingness is written: §F

`purity_pct` and `purity_method` currently hold the literal string TBD on all 65 oligos. §F requires NOT_REPORTED, and is explicit that it "is the correct value, not a failure state". TBD says *we have not got to it yet*; NOT_REPORTED says *the source does not report it*. For kidney the second is true and verified — I searched both source patents for purity/HPLC/UPLC/LC-MS language and neither reports any.

No Characterization Gap Register exists in toxicity/kidney. §F requires the value *and* the register. I have not renamed the values, because changing 130 cells in `data/oligos.csv` is a write to validated data and the standing rule gates that. **Proposed, not done**.

German's calibration — zero of 45 thrombo records with a usable purity value — matches kidney exactly at 0 of 65. This is a corpus property, not a kidney failure. Plan for disclosure.

Confirms work already done: §E "unreported is not negative"

`negative_eligibility` already implements this: five tiers, machine-derived, with `nephrotox_grade_modeling` left blank where a row is not eligible, so unsupported zeros become missing rather than zero. 20 rows are gated out. This rule arrived after the work and agrees with it.

No change: §D, §G, §H, §I

- **§D** — human clinical / human lab / animal are already separate lanes via `subject_class`, and animal evidence sits in appendix tabs. Kidney does not use GOLD/SILVER/BRONZE tiers at all.
- **§G** — **no model exists in** toxicity/kidney; nothing trains, splits or reports an AUC. The leakage rules bind the moment one does, and §C's two missing grouping fields are the blocker.
- **§H** — kidney is not named in the release table. One clarification to avoid a misreading: `scripts/release_check.py` prints `RELEASE CHECKS PASSED`. That is an **internal integrity gate** — schema, hashes, kidney-only isolation, tab order. It is **not** a dataset freeze and must not be read as one. German grants freezes; this script does not.
- **§I** — **none of the named corrections touch kidney**. Searched every `.csv`, `.md` and `.py` for Peacock, Goodchild, Hornung, Herzner, Riera-Tur, Fucini and Lenert: **zero files each**. Nothing to carry.

2. Grain declaration (§B) — flagged, nothing re-grained

Kidney is not one-row-per-oligo. Measured: 246 rows across 65 oligos, 3.8 rows per oligo, in the two-table architecture §B asks for — `data/oligos.csv` as the canonical identity/chemistry table, `data/measurements.csv` as the experimental-observation table, linked on `oligo_id`.

Tested for condition collapse on the key `oligo × species × system × dose × unit × duration × delivery × endpoint`: **246 distinct keys for 246 rows — zero duplicates**.

***CORRECTION, same day.** The sentence that followed this — "zero collapsed rows. No row stands for several doses or timepoints" — was wrong, and the test behind it was the wrong test.*

Key-uniqueness proves no two rows duplicate a condition; it says nothing about whether one row summarises many. A drug-level summary row has a unique key too. Re-measured properly:

- 37 of 65 oligos (57%) have exactly one row standing for their entire evidence.

- 98 of 246 rows cannot be resolved to a single dose, a single exposure time and a numeric value.

*- Kidney is **two regimes**: the patent panels are **129/150 (86%) resolvable** and fully crossed; everything else is **19/96 (20%)**.*

- 11 rows carry `species = multi_species`, pooling several species into one row; 21 carry an ordinal `in_vivo_nephrotoxicity_composite` bin rather than a measured quantity.

*So the honest answer to §B is: ****the patent-derived half is at condition grain; the rest is at drug-summary grain.**** Still not one-row-per-oligo overall, and still nothing re-grained — but the clean bill of health below was unearned. Detail in `ROCKSTEADY_TO_BEEBOP_CRANK_2026-10-03.md`.*

But the grain is incomplete against §B's full unit, and I would rather say so than claim compliance:

§B axis	state in kidney
oligo	present
chemistry	molecule-level only — no position encoding (§C above)
strand state	absent — no <code>strand_role</code> , no <code>duplex_partner_id</code>
dose	present, 39/246 rows (16%) have none
time	present, 20/246 (8%) have none
donor / cell system	system present; donor axis entirely absent — no <code>donor_id</code> , <code>donor_class</code> , <code>cell_subset</code> , <code>sample_state</code>
delivery condition	<code>delivery_method</code> present; no formulation or <code>delivery_agent</code>
endpoint	present, 246/246

So: **experiment-condition grain on the dose × time × system × endpoint axes, with the donor and strand axes missing and chemistry at the wrong level.** Re-graining is not the issue here — the rows are already per-condition. The issue is missing axes, which is a schema question for the Tier 0 crosswalk, not a re-graining question. **I have re-grained nothing.**

GSRS: kidney holds no GSRS data — searched, zero hits. The raw-and-unparsed staging rule has nothing to bind on here.

3. Delegated row — status

**Kidney: Correct the false "blocked" assertion at SOURCES.md:12 and SOURCE_REGISTER.md §2 ...*

*Do not act on the audit's Directive 5 ... MSR066 awaits German.**

a. False "blocked" assertion — corrected, and the correction paid for itself immediately.

Crank is right, and the mechanism is exactly as described. Measured on one URL:

user agent	result
default curl	HTTP 404, 420 bytes — an apology page that reads like a block
browser	HTTP 200, 7,558,885 bytes — the document

accessdata.fda.gov is **user-agent gated, never blocked**. Corrected at SOURCES.md:12 and in two places in SOURCE_REGISTER.md.

I then re-tested the rest of that same sentence, because a line that is wrong in one clause may be wrong in others. It is not: **NEJM, Circulation/AHA, ScienceDirect and academic.oup.com return 403 under both the default and a browser agent.** Those are genuine blocks and the claim stands for them. The correction is therefore narrow and specific rather than a blanket rewrite.

Acquired under the standing rule (acquisition proceeds now): four FDA Pharmacology/Toxicology reviews, 22.9 MB, previously recorded as requiring hand-delivery —

document	pages	renal content (first 40 pp)
inotersen, NDA 211172	134	kidney ×23, glomerular ×6, basophilic granulation ×5, creatinine ×4
golodirsen, NDA 211970	114	kidney ×34, renal ×29, basophilic granulation ×5, tubular ×22
casimersen, NDA 213026	76	kidney ×26, renal ×15, tubular ×28
viltolarsen, NDA 212154	49	kidney ×52, proximal tubule ×9, basophilic granulation ×6

All four are authoritative nonclinical renal toxicology for oligos already on our roster, and all four carry **basophilic granulation** — the classic ASO proximal-tubule finding — plus NOAELs. Staged in research_staging/papers/ with URL, date, HTTP code, byte size and SHA-256 in logs/acquisition_manifest.json. **Not ingested.**

A licensing note that differs from the rest of that folder: these are **US federal agency work products and therefore public domain**, consistent with Beebop's item 12 (the public-domain reading reaches US federal agencies only). They are excluded from git for size, not for rights. 217388 (eplontersen) and 219019 (nedosiran) were not located under the probed paths and remain open.

b. Directive 5 — verified, and not acted on. Crank's figures are exactly right:

```
staged rows in sequences_recovered_2026-10-02.csv : 10
oligos whose sequence is TBD                      : 10
INTERSECTION                                       : 6  <- not 10
staged ids that are NOT gaps                      : OLG002, OLG005, OLG006, OLG032
sequences now                                     : 55/65
if every staged row were promoted                 : 61/65  <- not 65/65
```

The four non-gap rows are exactly what my staging file labels them: two per-position chemistry upgrades for oligos that already have sequences (OLG006 mipomersen, OLG005 volanesorsen), one notation discrepancy for German

(OLG032 pelacarsen, U/T mixing), and one independent confirmation (OLG002 SPC5001). **An audit that counted ten staged rows as ten gap fills misread the file's own status column.** My Revision A reported 61 and so does this check; the two agree. Nothing promoted.

Residual after a full promotion would be 4: OLG014/OLG015 (Sandelius 2020, SAGE 403 on all routes) and OLG030/OLG031, which are **class-level aggregates and not molecules at all** — they can never carry a sequence and distort any coverage denominator that includes them. Coverage is better stated as **61/63 molecules** than 61/65 roster entries; both are true, and they answer different questions.

c. MSR066 — unchanged, awaiting German. Label untouched. §E now supplies a second, mechanical disqualification (§1 above) alongside the source-based one. German rules.

4. Every figure checked

Figure as given	Source	My measurement	Verdict
accessdata: 420-byte apology vs 2,181,318-byte PDF	Crank	420 B apology confirmed; browser PDF 7,558,885 B on NDA 211172	mechanism confirmed , my byte count differs — presumably a different document
Directive 5 id-set intersection = 6	Crank	6	confirmed
Promotion reaches 61/65, not 65/65	Crank	61/65	confirmed
Kidney has no control column	Beebop item 9	no control_role; only effect_vs_control, which is a comparator note, not a control-arm designation	confirmed
Canonical 246-row measurements.csv exists only on this branch	Crank	246 rows present here; cross-branch state not checked from this session	partially verified, not disputed
Kidney built one-row-per-oligo	implied by the all-endpoints flag	false for kidney — 3.8 rows/oligo, 246 unique condition keys	does not apply here

5. Open for others

German — unchanged from my 2026-10-02 report, plus three raised by the rules themselves:

- MSR066, now failing §E's clean-negative test on dose and denominator as well.
- The single 0–3 grade spanning clinical and in-vitro lanes** (§E), 148 of 246 rows affected.
- Ratification or voiding of the entire agent-assigned nephrotox_grade column** (§A).
- The renal-indication rule, which §E already names as "therapeutic correction is not toxicity".

Tier 0 crosswalk — the 24 absent §C fields, above all the seven position-chemistry fields and the two grouping fields. Kidney has per-position chemistry in hand for six constructs with nowhere to put it.

Oscar — whether purity_pct/purity_method may be rewritten TBD → NOT_REPORTED (130 cells in validated data, hence gated), and whether the Characterization Gap Register is mine to create or belongs in the crosswalk.

Not mine, flagged rather than assumed: 217388 and 219019 FDA reviews remain unlocated; the cross-branch stale-measurements.csv problem is listed under Cross-branch, not under Kidney, and I have not touched other branches.

6. Expected release-manifest drift — declared, not repaired

Correcting SOURCE_REGISTER.md drifted it away from the hash bound in RELEASE_MANIFEST.json (release_id kidney-224e9a2f7916). scripts/release_check.py now reports:

```
RELEASE BLOCKED – 1 hard failure(s)
- every file bound by the manifest matches its hash (1 drifted)
```

This is the gate working, not a fault. Checked file by file: all six data files — oligos.csv, measurements.csv, the merged view, the bridge, the clinical register and the workbook — hash **OK**. No data file moved.

Four bound **documents** drifted, each for a stated reason, all of them description:

file	why
SOURCE_REGISTER.md	the false accessdata.fda.gov "blocked" entry corrected
SOURCE_REGISTER.pdf	re-rendered so the PDF stops publishing the corrected-away claim
STATUS.md	§3a purity item partly overtaken — FDA reviews do report per-lot purity
schema.md	purity_pct note updated for the same reason

I did not regenerate the manifest. Doing so mints a new release_id, which is release mechanics and gated. The correct state is exactly what the gate now reports: the released artefacts no longer match their corrected sources, so **this release is not current** and a regeneration is owed once release is ungated. Anyone reading kidney-224e9a2f7916 should treat the source-access claims in it as superseded by the 2026-10-03 correction.

RULES READ AND APPLIED — ACQUISITION AND DESCRIPTION DONE, INGESTION AND PROMOTION HELD